



CYBERSECURITY TRAINING SESSION

Zero Trust Management

“Never Trust, Always Verify”



Identity

Device

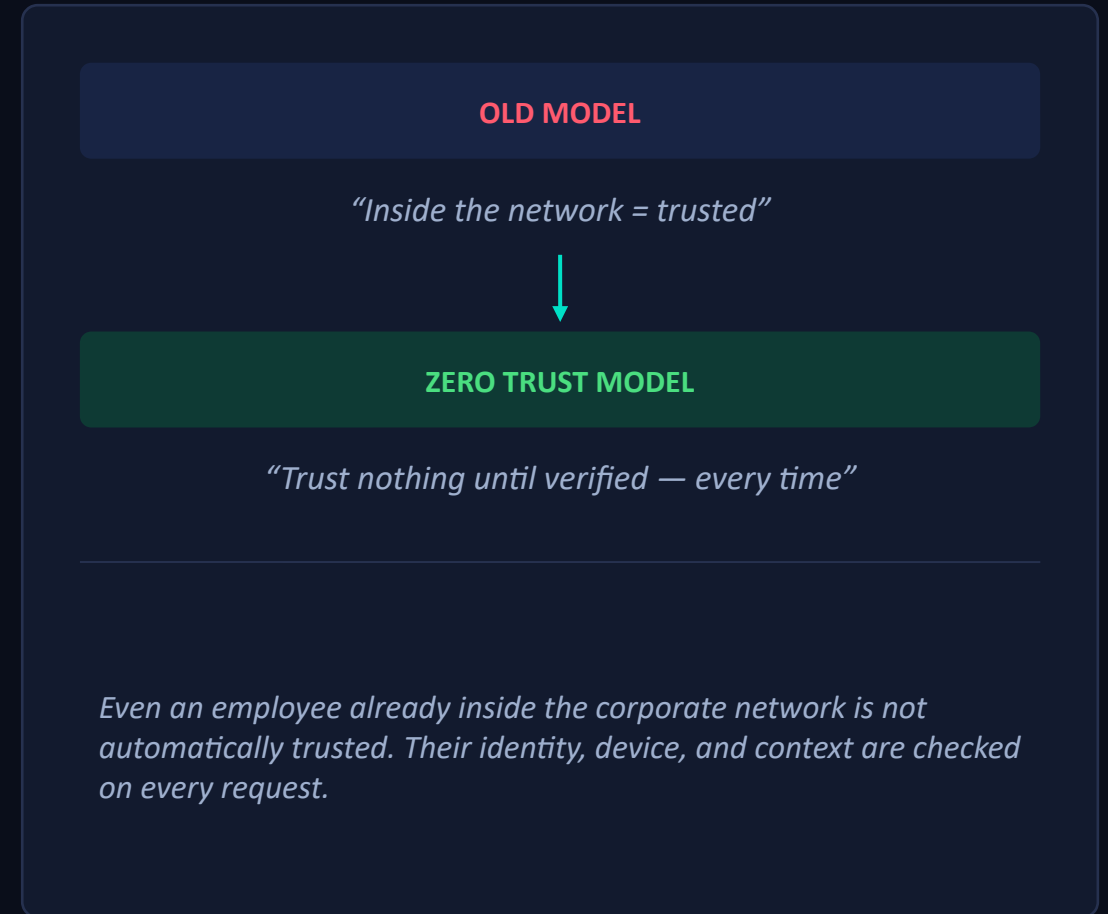
Network

Application

Data

What is Zero Trust?

- A security model built on one rule: never trust automatically
- No implicit trust for any user, device, or system
- Every access request is verified, regardless of location
- Validation happens continuously, not just at login
- Access is granted using least privilege



Why Do We Need Zero Trust?



Cloud Apps

SaaS platforms outside the traditional network



Remote Work

Employees connect from homes, cafes, airports



BYOD

Personal phones and laptops access company data



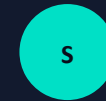
Mobile

Access happens anywhere, anytime, any device



Insider Risk

Threats can come from trusted accounts too



Stolen Creds

Passwords are phished, leaked, and reused

Traditional Security vs. Zero Trust

	TRADITIONAL PERIMETER	ZERO TRUST
Trust model	Trust inside the network	Verify continuously, everywhere
Access basis	Network location / perimeter	Identity-centric
Remote access	VPN-based access	Context-aware, per-request access
Controls	Static controls	Dynamic, least-privilege controls
After login	Broad access once authenticated	Continuous monitoring & re-checks

The Core Principle

“Never Trust, Always Verify”

- Every request is treated as potentially untrusted
- Authenticate — confirm who is asking
- Authorize — confirm what they may access
- Validate context — device, location, risk
- Monitor continuously — re-check throughout the session



A request only becomes access after passing every check.

Three Core Principles of Zero Trust

1

Verify Explicitly

Always authenticate and authorize using every available data point — identity, device, location, and behavior.

EXAMPLE

Login requires MFA plus a device-health check, not just a password.

2

Least Privilege Access

Give users only the access they need, limited in scope and, where possible, time.

EXAMPLE

A contractor gets read-only access to one project folder for 30 days.

3

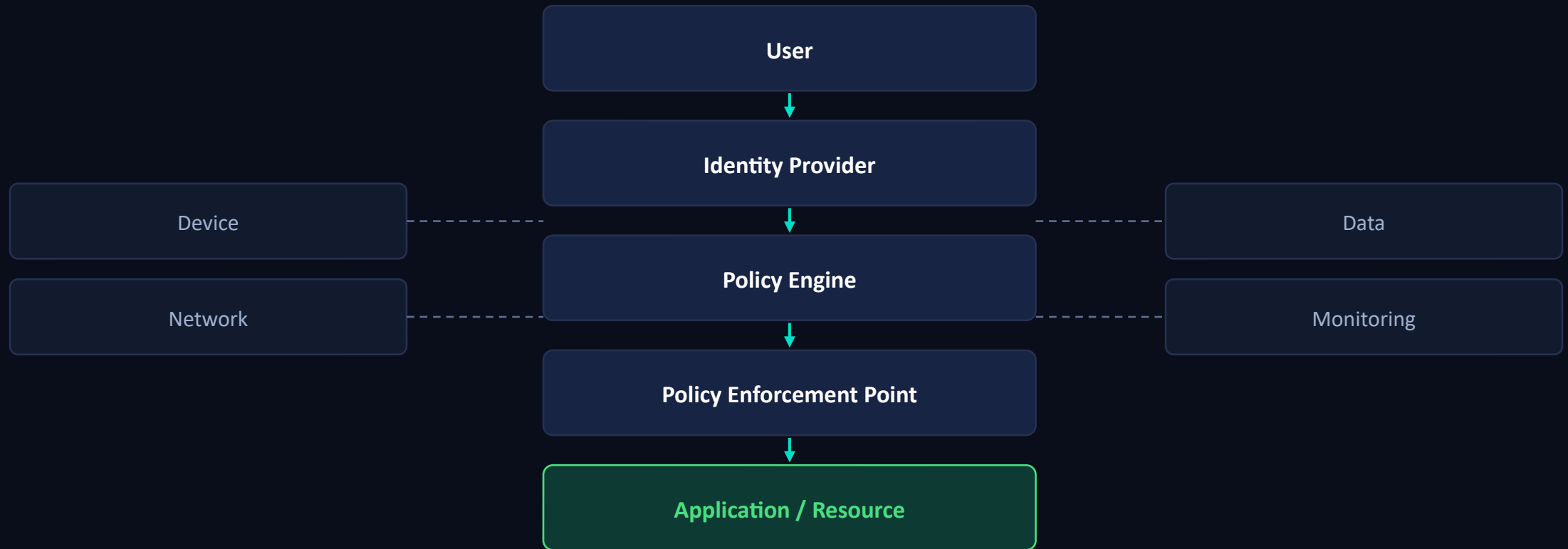
Assume Breach

Design as if an attacker is already inside — segment access and monitor continuously.

EXAMPLE

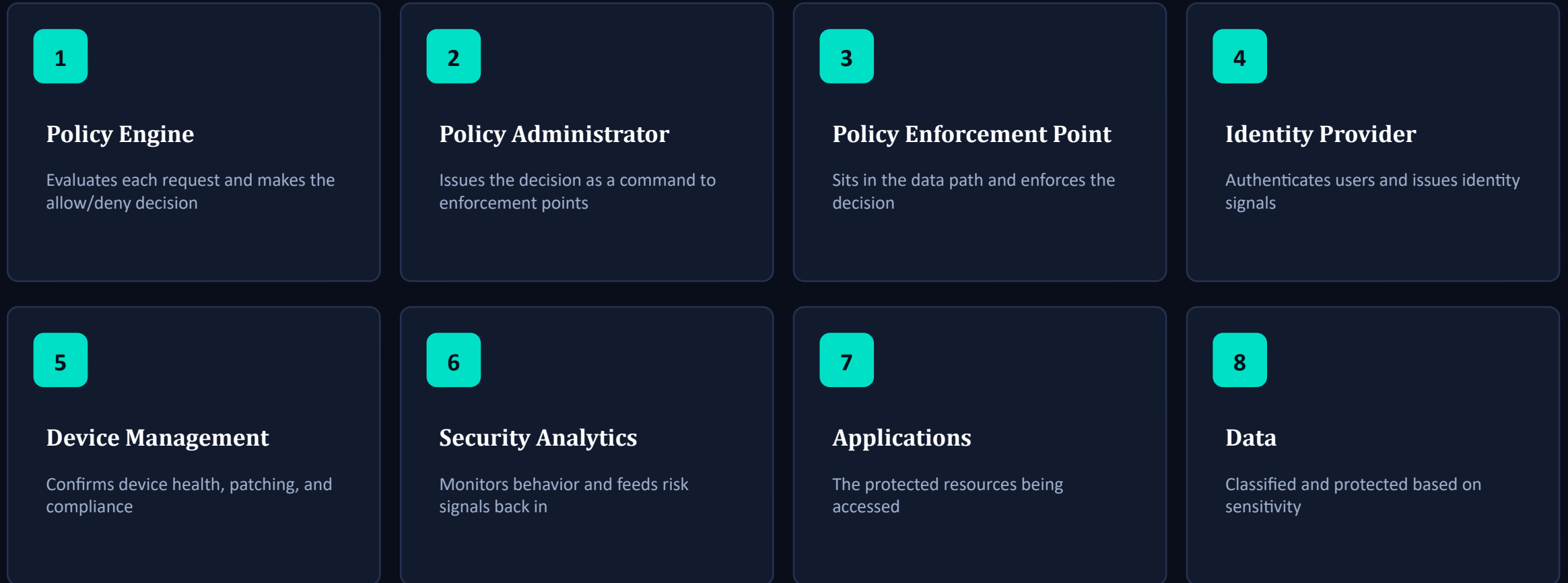
One compromised laptop cannot reach the entire internal network.

Zero Trust Architecture



Threat Intelligence feeds the Policy Engine with real-time risk signals.

Zero Trust Reference Architecture



Identity Is the New Perimeter

- User identity is the primary control point
- Strong authentication confirms who is asking
- MFA adds a second layer beyond passwords
- SSO simplifies secure access across apps
- IAM governs accounts, roles, and permissions
- Risk-based authentication adapts to context
- Privileged identities need extra scrutiny



Real-World Example

An employee logs in from a device that has never been seen before, from a country they have never traveled to.



Traditional model

Access likely allowed once password is correct



Zero Trust model

Risk score rises → step-up MFA is triggered

Authentication vs. Authorization

AUTHENTICATION

“Who are you?”

Username/password, MFA, biometrics — proving your identity.

AUTHORIZATION

“What can you access?”

RBAC and application permissions — defining allowed actions.

